

Supplementary Information

Individual infectiousness timing as a hidden driver of epidemic variability and superspreading

Carolyn Fulton, Liza Hadley, Casey Middleton, Stephen M. Kissler

Contents

1 Supplementary Methods

1.1	The Gamma time-shifted model: detailed specification	3
1.2	Contact process framework	3
1.3	Derivation of the overdispersion formula	4
1.3.1	Expressing k in terms of $\text{Var}[\tilde{c}_i]$	4
1.3.2	Deriving $\text{Var}[\tilde{c}_i]$ via spectral analysis	5
1.3.3	The power transfer function $ \hat{g}_i(\omega) ^2$	5
1.3.4	Overdispersion for the periodic contact model	6
1.3.5	Overdispersion for the stochastic contact model	6
1.4	Overdispersion in daily infection counts	6
1.5	Derivation of $E[\zeta]$ and $\text{Var}[\zeta]$: early-epidemic time shifts	7
1.6	Effective sample size for generation interval estimation	8
1.7	Invariance of infection-anchored TE to the shape of the infectiousness profile	9
1.8	Derivation of the truncated generation interval distribution under D&I interventions	9
1.9	Derivation of R_{eff} and OD for gathering size restrictions	9
1.10	Simulation framework	10
1.11	Benchmark pathogens	10
1.12	Empirical ψ estimation procedure	11
1.13	Power analysis for ψ inference	11

2 Supplementary Discussion

2.1	Why has ψ been overlooked?	11
2.2	Early-epidemic time shifts	12
2.3	Inference of r and $g(\tau)$	12
2.4	Detect-and-isolate interventions	12
2.5	Evaluating interventions beyond R_{eff}	12
2.6	Biases in empirical ψ estimates	13
2.7	Limitations	13

32	3	Supplementary Results	13
33	3.1	Detailed early-epidemic dynamics	13
34	3.2	Growth rate estimation uncertainty	14
35	3.3	Generation interval estimation and coverage	14
36	3.4	Detailed overdispersion results	14
37	3.5	Detailed D&I intervention results	14
38	3.6	Gathering size restriction results	14
39	4	Supplementary Figures	15
40	5	Supplementary Tables	15

Supplementary Methods

The Gamma time-shifted model: detailed specification

We seek an analytically tractable definition of $g_i(\tau)$ such that $g(\tau) = E_i[g_i(\tau)]$, and where the width of $g_i(\tau)$ is governed by a single smoothness parameter $\psi \in [0, 1]$ that interpolates between the “spike” and “smooth” extremes.

We first assume that $g(\tau)$ follows a $\text{Gamma}(\alpha, \beta)$ distribution with rate $\beta = \alpha/\bar{g}$, where $\bar{g}$ is the mean generation time. Next, we define each person’s individual intrinsic generation interval distribution as

$$g_i(\tau) = \begin{cases} 0 & \tau \leq l_i \\ f_\epsilon(\tau - l_i) & \tau > l_i \end{cases} \quad (1)$$

where $l_i \sim \text{Gamma}((1 - \psi)\alpha, \beta)$ is a random latent period and f_ϵ is the $\text{Gamma}(\psi\alpha, \beta)$ density. Equivalently, the secondary infections j from an index case i are distributed at times $\tau_{ij} = l_i + \epsilon_j$, where $l_i \sim \text{Gamma}((1 - \psi)\alpha, \beta)$ and $\epsilon_j \stackrel{iid}{\sim} \text{Gamma}(\psi\alpha, \beta)$.

Since l_i and ϵ_j are independent Gamma-distributed random variables with common rate β , their sum follows a $\text{Gamma}((1 - \psi)\alpha + \psi\alpha, \beta) = \text{Gamma}(\alpha, \beta)$ distribution; thus $E_i[g_i(\tau)] = g(\tau)$, as required. Furthermore, when $\psi \rightarrow 1$, the latent period vanishes ($l_i \rightarrow 0$) and f_ϵ approaches the full $\text{Gamma}(\alpha, \beta)$ density, which gives the “smooth” extreme (no between-individual variation). Similarly, when $\psi \rightarrow 0$, f_ϵ concentrates to a point mass and $g_i(\tau)$ becomes a delta function at $l_i \sim \text{Gamma}(\alpha, \beta)$, which gives the “spike” extreme (no within-individual variation).

We generally assume that ψ is a single value shared by all individuals for a given pathogen, but the framework is more general: the Gamma addition property ensures that drawing individual ψ_i from a distribution on $[0, 1]$ still yields $E_i[g_i(\tau)] = g(\tau)$; thus, it is possible to model scenarios where individuals have a mixture of narrow and broad individual infectiousness profiles.

Contact process framework

Under time-uniform contacts, $a_i(\tau) = \nu_i g_i(\tau)$, where ν_i and $g_i(\tau)$ are defined with respect to a reference contact level which we denote $\bar{c} \equiv 1$. To allow for time-varying contacts, we introduce a contact process $\mathbf{c}(t)$: a nonnegative, wide-sense stationary stochastic process with time-mean $E_t[\mathbf{c}(t)] = 1$. Each person i has a contact modulator $c_i(t)$ that is a realization of $\mathbf{c}(t)$, representing the proportional scaling of that person’s contact rate relative to the reference level at calendar time t . The constraint $E_t[\mathbf{c}(t)] = 1$ is an ensemble average over all possible realizations at time t ; individual realizations c_i need not have time-average 1, allowing for individuals with persistently higher or lower contact rates than the population average. We additionally define the “biological infectiousness”, b_i , as the expected number of secondary infections person i would produce under $\bar{c} \equiv 1$. The individual infectiousness profile thus generalizes to

$$a_i(t_i, \tau) = b_i g_i(\tau) c_i(t_i + \tau) \quad (2)$$

71 where t_i is person i 's infection time. If $c(t) \equiv \bar{c} = 1$, this reduces back to $a_i(\tau) = v_i g_i(\tau)$, with $b_i = v_i$. More
 72 generally,

$$v_i = b_i \int_0^\infty g_i(\tau) c_i(t_i + \tau) d\tau \quad (3)$$

73 For the population-level quantities,

$$R_0 = E_i[v_i] = E_i[b_i] \quad \text{and} \quad g(\tau) = E_i[b_i g_i(\tau)] / R_0 \quad (4)$$

74 where the second equality follows from the assumed independence of (b_i, g_i) and c_i and the constraint
 75 $E[c(t)] = 1$.

76 **Periodic contact model.** We define the “periodic” contact model

$$c_i(t) = 1 - c_{\text{amp}} \cos\left(\frac{2\pi t}{c_{\text{per}}}\right) \quad \forall i \quad (5)$$

77 with amplitude $c_{\text{amp}} \in [0, 1]$ and period $c_{\text{per}} > 0$. In this model, all individuals share the same deterministic
 78 contact function; while there is temporal variation within each person's contacts, there is no variation across
 79 individuals. If infection times t_i are approximately uniformly distributed over one period — as may hold
 80 during exponential growth when c_{per} is short relative to the epidemic timescale — then the effective contact
 81 level experienced by a randomly chosen individual at time τ after infection averages to 1.

82 **Stochastic contact model.** We also define the “stochastic” contact model, where each person's $c_i(t)$ is a
 83 piecewise-constant process that switches to a new level at arrivals of a Poisson process with rate λ . The
 84 contact levels are drawn independently from a $\text{Gamma}(\sigma, \sigma)$ distribution, which has mean 1 and variance
 85 $1/\sigma$. This creates a two-parameter family indexed by the dispersion parameter $\sigma > 0$ and switching rate
 86 λ . Smaller σ yields more heterogeneous contacts. In contrast to the periodic model, each person's contact
 87 trajectory is unique, but the population average is constant in time: $C(t) = E_i[c_i(t)] = 1$.

88 Derivation of the overdispersion formula

89 *Expressing k in terms of $\text{Var}[\tilde{c}_i]$*

90 We begin with the standard assumption that the number of offspring Z_i produced by an index case i is
 91 Poisson-distributed:

$$Z_i \sim \text{Poisson}(v_i) \quad (6)$$

92 When v_i is uniform across people ($v_i \equiv R_0$), then the Z_i are truly Poisson-distributed. Overdispersion occurs
 93 when v_i varies across individuals. To capture this overdispersion, the standard approach is to describe Z_i
 94 using a Negative Binomial distribution:

$$Z_i \sim \text{NegBin}(R_0, k) \quad (7)$$

95 which has expectation R_0 and variance $R_0(1 + R_0/k)$. The parameter k can be obtained *via* moment match-
 96 ing:

$$k = \frac{R_0^2}{\text{Var}[Z_i] - R_0} \quad (8)$$

97 Furthermore, since

$$\text{Var}[Z_i] = \text{Var}[v_i] + E[v_i] = \text{Var}[v_i] + R_0 \quad (9)$$

98 the expression simplifies to

$$k = \frac{R_0^2}{\text{Var}[v_i]} \quad (10)$$

99 If everyone's inherent biological infectiousness is the same ($b_i \equiv R_0$), we have

$$v_i = R_0 \int_0^\infty g_i(\tau) c_i(t_i + \tau) d\tau := R_0 \tilde{c}_i \quad (11)$$

100 and thus

$$k = \frac{1}{\text{Var}[\tilde{c}_i]} \quad (12)$$

101 ***Deriving $\text{Var}[\tilde{c}_i]$ via spectral analysis***

The random variable $\tilde{c}_i$ inherits randomness from both $g_i(\tau)$ and $c_i(t_i + \tau)$. By the Law of Total Variance:

$$\text{Var}[\tilde{c}_i] = \text{Var}[E[\tilde{c}_i|g_i]] + E[\text{Var}[\tilde{c}_i|g_i]] \quad (13)$$

$$= \text{Var}[1] + E[\text{Var}[\tilde{c}_i|g_i]] = E[\text{Var}[\tilde{c}_i|g_i]] \quad (14)$$

102 Conditional on g_i , the quantity $\tilde{c}_i|g_i = \int_0^\infty g_i(\tau) c_i(t_i + \tau) d\tau$ has the form of a classic linear system
 103 problem: c_i is a wide-sense stationary stochastic process (input signal), g_i is an impulse response, and $\tilde{c}_i|g_i$
 104 is the output evaluated at time t_i . Following standard signal processing theory:

$$\text{Var}[\tilde{c}_i|g_i] = R_{\tilde{c}_i|g_i}(0) - 1 \quad (15)$$

105 where $R_\bullet(\tau)$ denotes the autocorrelation function. In frequency space:

$$R_{\tilde{c}_i|g_i}(0) = \frac{1}{2\pi} \int_{-\infty}^\infty S_{\tilde{c}_i|g_i}(\omega) d\omega \quad (16)$$

106 and by the fundamental theorem for linear systems:

$$S_{\tilde{c}_i|g_i}(\omega) = |\hat{g}_i(\omega)|^2 S_{c_i}(\omega) \quad (17)$$

107 ***The power transfer function $|\hat{g}_i(\omega)|^2$***

Under the Gamma time-shift model, $g_i(\tau) = f_\epsilon(\tau - l_i)$. Taking the Fourier transform:

$$\hat{g}_i(\omega) = \int_{l_i}^\infty f_\epsilon(\tau - l_i) e^{-i\omega\tau} d\tau = e^{-i\omega l_i} \left(\frac{\beta}{\beta + i\omega} \right)^{\psi\alpha} \quad (18)$$

108 Taking the squared modulus:

$$|\hat{g}_i(\omega)|^2 = \left(1 + \frac{\omega^2}{\beta^2} \right)^{-\psi\alpha} \quad (19)$$

109 This is constant across individuals and does not depend on l_i .

110 **Overdispersion for the periodic contact model**

111 For the periodic contact model $c_i(t) = 1 - c_{\text{amp}} \cos(\omega_0 t)$ with $\omega_0 = 2\pi/c_{\text{per}}$, the power spectral density is:

$$S_{c_i}(\omega) = 2\pi\delta(\omega) + \frac{\pi c_{\text{amp}}^2}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \quad (20)$$

Assembling:

$$\begin{aligned} \text{Var}[\tilde{c}_i|g_i] &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(1 + \frac{\omega^2}{\beta^2}\right)^{-\psi\alpha} \left[2\pi\delta(\omega) + \frac{\pi c_{\text{amp}}^2}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \right] d\omega - 1 \\ &= \frac{c_{\text{amp}}^2}{2} \left(1 + \frac{\omega_0^2}{\beta^2}\right)^{-\psi\alpha} \end{aligned} \quad (21)$$

112 Thus,

$$k = \frac{2}{c_{\text{amp}}^2} \left(1 + \left(\frac{2\pi\bar{g}}{\alpha c_{\text{per}}}\right)^2\right)^{\psi\alpha} \quad (22)$$

113 or equivalently, $\text{OD} \equiv 1/k = \frac{c_{\text{amp}}^2}{2} \left(1 + \left(\frac{2\pi\bar{g}}{\alpha c_{\text{per}}}\right)^2\right)^{-\psi\alpha}$.

114 **Overdispersion for the stochastic contact model**

115 For the stochastic contact model with $\text{Gamma}(\sigma, \sigma)$ -distributed levels switching at rate λ , the autocorrelation is:

$$R_{c_i}(\tau) = \frac{1}{\sigma} e^{-\lambda|\tau|} + 1 \quad (23)$$

117 and the power spectral density is:

$$S_{c_i}(\omega) = \frac{2\lambda}{\sigma(\lambda^2 + \omega^2)} + 2\pi\delta(\omega) \quad (24)$$

118 Assembling:

$$\text{Var}[\tilde{c}_i|g_i] = \frac{1}{\sigma} \int_{-\infty}^{\infty} \frac{\lambda}{\pi(\lambda^2 + \omega^2)} \left(1 + \frac{\omega^2}{\beta^2}\right)^{-\psi\alpha} d\omega \quad (25)$$

119 and k is the reciprocal of this integral. While the integral can be expressed in terms of special functions, we
120 evaluate it numerically in practice.

121 **Overdispersion in daily infection counts**

122 Here we derive the index of dispersion of the daily infection count attributable to a single infector.

123 **Setup.** Fix an index case i , infected at time t_i , who generates $\chi_i \sim \text{Poisson}(R_0)$ total secondary infections.

124 Let $N_i(\tau)$ be the number of person i 's secondary infections that fall on day-since-infection $[\tau, \tau + 1)$, and

125 define the random landing probability

$$Q_i(\tau) = \int_{\tau}^{\tau+1} f_{\epsilon}(u - l_i) du \quad (26)$$

126 **Mean of $Q_i(\tau)$.** Taking the expectation over l_i :

$$E[Q_i(\tau)] = \int_{\tau}^{\tau+1} (f_l * f_{\epsilon})(u) du = \int_{\tau}^{\tau+1} g(u) du \equiv q(\tau) \quad (27)$$

127 which is independent of ψ .

128 **Index of dispersion.** By the Poisson thinning property, $N_i(\tau) \mid Q_i(\tau) \sim \text{Poisson}(R_0 Q_i(\tau))$. The law of
129 total variance gives:

$$\text{ID}[N_i(\tau)] = \frac{\text{Var}[N_i(\tau)]}{E[N_i(\tau)]} = 1 + R_0 \frac{\text{Var}[Q_i(\tau)]}{E[Q_i(\tau)]} \quad (28)$$

130 In the smooth case ($\psi = 1$), $Q_i(\tau) = q(\tau)$ is deterministic, so $\text{ID}[N_i(\tau)] = 1$. In the spike case ($\psi = 0$),
131 $Q_i(\tau) = \mathbf{1}(l_i \in [\tau, \tau + 1))$ is Bernoulli with parameter $q(\tau)$, giving $\text{ID}[N_i(\tau)] = 1 + R_0(1 - q(\tau))$.

132 **Derivation of $E[\zeta]$ and $\text{Var}[\zeta]$: early-epidemic time shifts**

133 **Setup.** Following Morris et al. [2024], we describe the cumulative infections during the early-epidemic
134 exponential growth phase as a branching process, $Z(t)$. Let r be the epidemic's exponential growth rate;
135 then, as $t \rightarrow \infty$, $e^{-rt}Z(t) \rightarrow W$ almost surely, where W is a random variable capturing stochastic fluctuations
136 in the early-epidemic process. With a change of variables, we translate W into a random initial time shift:

$$Z(t) \approx e^{r(t+\zeta)} \quad \text{where} \quad \zeta = \frac{\log W}{r} \quad (29)$$

137 when $I_0 = E[W] = 1$. Our goal is to determine how $E[\zeta]$ and $\text{Var}[\zeta]$ depend on ψ .

138 **Approximating the distribution of W .** The distribution of W consists of a point mass at 0 with mass q
139 (the probability of epidemic extinction), combined with a smooth density with positive support. We restrict
140 attention to $W|\text{surv}$, and approximate it with a $\text{Gamma}(k, \lambda)$ distribution matched to the first two moments:

$$k = \frac{E[W|\text{surv}]^2}{\text{Var}[W|\text{surv}]} \quad \text{and} \quad \lambda = \frac{E[W|\text{surv}]}{\text{Var}[W|\text{surv}]} \quad (30)$$

141 **Deriving $E[W|\text{surv}]$.** Since $E[W|\text{extinct}] = 0$:

$$E[W|\text{surv}] = \frac{E[W]}{p} = \frac{1}{p} \quad (31)$$

142 where $p = 1 - q$ is the probability of epidemic survival.

143 **Deriving $\text{Var}[W]$.** We introduce the notation $\rho_c = 1/(1 + cz)$, where $z = r/\beta = R_0^{1/\alpha} - 1$ (from the
144 Euler–Lotka equation). Using the distributional fixed-point equation $W \stackrel{d}{=} \sum_{j=1}^{\chi_i} e^{-r\tau_j} W_j$ and the law of total

145 variance:

$$\text{Var}[W] = \frac{\text{Var}[\sum_j e^{-r\tau_j}]}{1 - R_0 \rho_2^\alpha} \quad (32)$$

146 Since $\tau_j = l_i + \epsilon_j$, and $l_i, \epsilon_j, \epsilon_k$ are independent:

$$E[e^{-r\tau_j} e^{-r\tau_k}] = E[e^{-2rl_i}](E[e^{-r\epsilon}])^2 = \rho_2^{(1-\psi)\alpha} \cdot \rho_1^{2\psi\alpha} \quad (33)$$

147 Assembling and substituting $\sigma_c = (1+z)^c / (1+2z)$:

$$\text{Var}[W] = \frac{\sigma_1^\alpha + \sigma_2^{(1-\psi)\alpha} - 1}{1 - \sigma_1^\alpha} \quad (34)$$

148 The conditional variance is then:

$$\text{Var}[W|\text{surv}] = \frac{1}{p}(1 + \text{Var}[W]) - \frac{1}{p^2} \quad (35)$$

149 **Parameters of the moment-matched Gamma.** Combining gives:

$$k = \frac{1 - \sigma_1^\alpha}{p\sigma_2^{(1-\psi)\alpha} - 1 + \sigma_1^\alpha}, \quad \lambda = pk \quad (36)$$

Mode, mean, and variance of ς .

$$\text{Mode}[\varsigma] = \frac{1}{r} \log \frac{1}{p} \quad (37)$$

$$E[\varsigma] = \frac{\log(1/p)}{r} + \frac{\Psi^{(0)}(k) - \log(k)}{r} \quad (38)$$

$$\text{Var}[\varsigma] = \frac{1}{r^2} \Psi^{(1)}(k) \quad (39)$$

150 where $\Psi^{(0)}$ is the digamma function and $\Psi^{(1)}$ is the trigamma function. The mode is independent of ψ . The
 151 mean shift $(\Psi^{(0)}(k) - \log k)/r$ is strictly negative (by Jensen's inequality) and strictly increasing in ψ (since
 152 k increases in ψ and $\Psi^{(0)}(k) - \log k$ increases in k). The variance $\Psi^{(1)}(k)/r^2$ is strictly decreasing in k , and
 153 thus strictly increasing as $\psi \rightarrow 0$.

154 In summary: punctuated profiles ($\psi \rightarrow 0$) produce more variable epidemic timing (higher $\text{Var}[\varsigma]$) and
 155 modestly later epidemics on average (more negative $E[\varsigma]$).

156 **Effective sample size for generation interval estimation**

157 When $\psi < 1$, serial intervals from the same index case are correlated (they share the latent period l_i).
 158 Under the time-shifted Gamma model, and assuming an independent Gamma($a_{\text{obs}}, b_{\text{obs}}$) observation delay

159 applied to each infection time, the intraclass correlation coefficient is:

$$\text{ICC} = \frac{(1 - \psi)\alpha/\beta^2 + (a_{\text{obs}}/b_{\text{obs}}^2)^2}{\alpha/\beta^2 + 2(a_{\text{obs}}/b_{\text{obs}}^2)^2} \quad (40)$$

160 Kish's design effect for cluster sampling gives an effective sample size:

$$n_{\text{eff}} = \frac{n}{1 + (R_0 - 1) \cdot \text{ICC}} \quad (41)$$

161 Thus, n_{eff} is minimized when $\psi \rightarrow 0$: when individual infectiousness profiles are narrow, the offspring
162 of an index case reflect effectively just one draw from $g(\tau)$, regardless of how many secondary infections
163 occur.

164 [Full derivation of the ICC formula to be added.]

165 **Invariance of infection-anchored TE to the shape of the infectiousness profile**

166 When detection timing is anchored to the infection time (*e.g.*, the detectability window spans a fixed interval
167 after infection), testing effectiveness depends only on $g(\tau)$ and not on ψ . This is because:

$$\text{TE} = \rho\eta P(\tau > D) = \rho\eta \int_D^\infty g(\tau) d\tau \quad (42)$$

168 which involves only the population-level $g(\tau)$. Since $g(\tau)$ is invariant to ψ by construction, so is TE under
169 infection-anchored detection.

170 [Full proof to be added.]

171 **Derivation of the truncated generation interval distribution under D&I interventions**

172 Under a detect-and-isolate intervention with mode-anchored detection, the effective generation interval
173 distribution becomes:

$$g^*(\tau) = \frac{(1 - \eta\rho)g(\tau) + \eta\rho \int_0^{\min(\tau, D)} f_I(\tau - \epsilon) f_\epsilon(\epsilon) d\epsilon}{1 - \text{TE}} \quad (43)$$

174 For smooth profiles ($\psi \rightarrow 1$), detection reliably removes late transmission, shortening the mean gen-
175 eration interval. For punctuated profiles ($\psi \rightarrow 0$), detection is all-or-nothing: among those who escape
176 detection, $g^*(\tau) = g(\tau)$ with no truncation.

177 [Full derivation to be added.]

178 **Derivation of R_{eff} and OD for gathering size restrictions**

179 When the stochastic contact distribution is truncated at a maximum gathering size c_{max} , the effective repro-
180 duction number is:

$$R_{\text{eff}} = \mu_T R_0 \quad \text{where} \quad \mu_T = \frac{F_{\text{Gamma}}(c_{\text{max}}; \sigma + 1, \sigma)}{F_{\text{Gamma}}(c_{\text{max}}; \sigma, \sigma)} \quad (44)$$

181 where $F_{\text{Gamma}}(c; a, b)$ is the $\text{Gamma}(a, b)$ CDF evaluated at c . This factor is independent of ψ .

The overdispersion changes as:

$$\frac{\text{OD}_{\text{new}}}{\text{OD}_{\text{old}}} = \frac{\sigma}{\mu_T^2 / V_T} \quad (45)$$

where V_T is the variance of the truncated Gamma(σ, σ) distribution. The relative change in OD is independent of ψ , but the absolute change is ψ -dependent through OD_{old} .

[Full derivation to be added.]

Simulation framework

We used stochastic, individual-based epidemic simulations to investigate the impact of individual-level infectiousness timing on epidemic dynamics. The simulation model has a modular design: an infection attempt generator produces a set of transmission attempts for each infected individual, and a population-level simulation script processes these attempts to propagate the epidemic. We implemented two simulation variants: a finite-population model with susceptible depletion, and an infinite-population model where every infection attempt succeeds. Both are implemented in R (version 4.5.0), with performance-critical inner loops in C++ via Rcpp.

Infection attempt generation. For each newly infected individual i (with infection time t_i), the generator:

1. Draws $\chi_i \sim \text{Poisson}(R_0)$ secondary infection attempts (with $b_i \equiv R_0$ throughout).
2. Draws a latent period $l_i \sim \text{Gamma}((1 - \psi)\alpha, \beta)$.
3. For each attempt j , draws an independent jitter $\epsilon_j \sim \text{Gamma}(\psi\alpha, \beta)$. The attempt occurs at time $\tau_{ij} = l_i + \epsilon_j$.
4. When a time-varying contact process is active, thins each attempt via Poisson thinning: accept with probability $c_i(t_i + \tau_{ij}) / c_{\text{max}}$.
5. When a detect-and-isolate intervention is active, draws a detection time D_i and removes post-detection attempts with probability η .

Finite-population simulation. Tracks susceptible depletion in a population of size N . Events are processed in chronological order from a priority queue. Each attempt targets a uniformly chosen individual; if susceptible, infection occurs.

Infinite-population simulation. A branching process approximation: every infection attempt succeeds, with no susceptibility check.

Default parameters. Unless otherwise stated: $N = 10,000$, $n_{\text{sim}} = 5,000$ replicates per scenario, epidemic establishment defined as reaching $0.05N = 500$ cumulative infections.

Benchmark pathogens

Pathogen	R_0	$\bar{g}$ (days)	$\text{Var}[g]$ (days ²)	α	β	Reference
Influenza	2	3.2	4.41	2.32	0.73	Chan et al. [2025]
SARS-CoV-2 (omicron)	6	7.05	20.8	2.39	0.34	Manica et al. [2022]
Measles	12	12.2	13.10	11.36	0.93	Klinkenberg and Nishiura [2011]

Table 1: Benchmark pathogen parameters used in simulations.

Empirical ψ estimation procedure

Transmission trees were obtained from the OutbreakTrees database[?], supplemented with COVID-19 data from ? and ?. Serial intervals were grouped by index case into clusters, restricting to clusters with ≥ 2 secondary cases (singleton clusters carry no information about within-cluster correlation).

For each pathogen, the generation interval distribution $g(\tau) \sim \text{Gamma}(\alpha, \beta)$ was fixed from published estimates. A flat (uniform) prior was placed on $\psi \in [0, 1]$, evaluated on a grid of 101 equally spaced points. For each cluster c containing n_c secondary infections with serial intervals $\{s_{c1}, \dots, s_{cn_c}\}$, the likelihood was computed by integrating over the shared latent period l :

$$\mathcal{L}_c(\psi) = \int_0^\infty f_l(l; (1 - \psi)\alpha, \beta) \prod_{j=1}^{n_c} f_\epsilon(s_{cj} - l; \psi\alpha, \beta) dl \quad (46)$$

where f_l and f_ϵ are the Gamma densities for the latent period and jitter, respectively. The total likelihood is $\mathcal{L}(\psi) = \prod_c \mathcal{L}_c(\psi)$.

Posterior mode, mean, and 95% credible intervals were computed from the normalized posterior on the grid. Integration over l was performed numerically.

[Table of fixed GI parameters per pathogen to be added.]

Power analysis for ψ inference

We simulated serial interval data for 10, 25, 50, and 100 index cases at $\psi \in \{0, 0.5, 1\}$ for each benchmark pathogen. For each infection, a Gamma-distributed observation delay was added, with mean delays of 1 day (fast), 2 days (medium), and 4 days (slow). Posterior estimates for ψ were generated using the procedure described above. Power was defined as $> 80\%$ of 200 replicates assigning $\geq 95\%$ posterior mass within 0.2 of the true ψ .

Supplementary Discussion

Why has ψ been overlooked?

A natural question is why the shape of the individual infectiousness profile has received so little attention. One reason is that ψ does not affect mean-field dynamics, by design: $E_i[g_i(\tau)] = g(\tau)$ regardless of ψ , so standard renewal equation models are ψ -invariant. Consequently, epidemic forecasts and retrospective analyses that rely on the renewal equation can proceed without specifying ψ , and for many purposes this is adequate.

A second reason is that the individual infectiousness profile has been dealt with implicitly in many set-

tings, without being isolated as an independent axis of variation. Individual-based models routinely make assumptions about $a_i(\tau)$'s shape — whether through SEIR-type stepwise profiles, viral kinetics-informed curves, or agent-based transmission rules — but these assumptions are typically adopted for modeling convenience rather than studied as objects of interest. Similarly, the linear chain trick can be used to narrow the individual infectiousness profile by stringing together sequences of exposed and infectious compartments, but this simultaneously changes $g(\tau)$, conflating two distinct effects[Wearing et al., 2005, Lloyd, 2001]. The central contribution of the present work is to decouple these effects.

Early-epidemic time shifts

For the early-epidemic time shift, the shift in $E[\zeta]$ is modest; the main effect is the inflation of $\text{Var}[\zeta]$, which widens the distribution of possible epidemic trajectories. This variability is relevant for scenario planning: early in an emerging epidemic, punctuated profiles make it harder to distinguish a “slow” epidemic from a “late” one, complicating early assessments of epidemic severity and the decision of when to intervene.

Inference of r and $g(\tau)$

The increased overdispersion in daily case counts and the reduced effective sample size of serial interval data both point to the need for statistical methods that account for within-cluster correlation. Treating serial intervals as independent draws from $g(\tau)$ — as is common practice — can produce overconfident estimates of α and β when individual infectiousness profiles are narrow. This suggests that contact tracing protocols should, where possible, record the identity of the index case for each observed serial interval, enabling cluster-level analyses that correctly account for the correlation structure.

Detect-and-isolate interventions

The sensitivity of testing effectiveness to ψ has practical implications. Shifts in symptom timing relative to peak infectiousness are commonplace — as occurred across SARS-CoV-2 variants — and changes in testing frequency are routine operational decisions. Our results indicate that the epidemic consequences of such shifts depend on the shape of the individual infectiousness profile: for punctuated profiles, small changes in detection timing can produce sharp, nonlinear changes in testing effectiveness, generation interval truncation, and overdispersion. This sensitivity underscores the importance of characterizing $a_i(\tau)$ for operational planning.

Additionally, the differential response of testing effectiveness to detection timing across ψ -values suggests that intervention data could, in principle, provide information about the individual infectiousness profile beyond what is available from contact tracing alone — an idea we have not developed here but consider a promising direction.

Evaluating interventions beyond R_{eff}

The effects of detect-and-isolate interventions and gathering size restrictions on epidemic dynamics extend beyond their impact on R_{eff} . Detect-and-isolate interventions simultaneously reduce R_{eff} , truncate the generation interval (accelerating transmission among those who escape detection), and modulate overdispersion — and these effects can work in opposing directions. Gathering size restrictions reduce R_{eff} but

also reduce overdispersion, which can lower epidemic extinction probabilities by a different amount than what the R_{eff} reduction alone would imply. Evaluating interventions purely in terms of their effect on R_{eff} can therefore be misleading; the framework introduced here provides a more complete accounting.

Biases in empirical ψ estimates

Several biases in the available data are likely to push $\hat{\psi}$ downward. Contact tracing studies preferentially capture large clusters and superspreading events, which can arise from event-based transmission (*e.g.*, a gathering at which many people are simultaneously exposed) rather than from a genuinely narrow infectiousness window; this ascertainment bias is particularly concerning for measles and pneumonic plague, where transmission is often event-mediated and well-traced. Incomplete observation of an infector’s full set of secondary infections can reduce within-cluster variance, mimicking low ψ . Date discretization to symptom onset day compresses within-cluster variance further. Publication bias in the OutbreakTrees database may over-represent notable outbreaks featuring unusual clustering.

Conversely, some biases push $\hat{\psi}$ upward: uncertainty in transmission direction can corrupt cluster assignments, and observation noise from variable incubation periods inflates apparent within-cluster variance. Additionally, our inference conditions on published generation interval parameters, introducing sensitivity to those estimates.

Local susceptible depletion could also produce patterns that mimic low ψ : for a highly infectious pathogen like measles, an index case may rapidly exhaust local susceptibles upon reaching peak infectiousness, yielding tightly clustered secondary infections even if the biological window of infectiousness is broad. Disentangling such epidemiological effects from the intrinsic shape of $a_i(\tau)$ would require data from settings where susceptible depletion is minimal, such as controlled exposure studies or outbreaks in large, well-mixed populations.

Limitations

The Gamma time-shifted model, while flexible, is only one possible parameterization of the within- *vs.* between-individual decomposition. Other families of distributions, or nonparametric approaches, might capture features that the Gamma model cannot. The contact process models we employ — periodic and stochastic — are useful for developing intuition but are difficult to calibrate empirically; real contact patterns are structured in ways (*e.g.*, household *vs.* community *vs.* workplace contacts) that these simple models do not capture. Our treatment of detect-and-isolate interventions, while covering several detection mechanisms, does not address the full range of possibilities (*e.g.*, contact tracing-initiated testing, adaptive screening). And our empirical inference of ψ is intentionally simplified: a more rigorous approach would jointly estimate ψ and the generation interval parameters, account for observation noise and ascertainment explicitly, and propagate uncertainty in incubation period estimates.

Supplementary Results

Detailed early-epidemic dynamics

We simulated 5,000 epidemics in a population of size 10,000 for each benchmark pathogen, varying $\psi \in \{0, 0.5, 1\}$, and measured the time to reach 5% cumulative prevalence (500 total infections). By day 20, only

311 55% of simulated SARS-CoV-2 epidemics with $\psi = 0$ had reached 5% prevalence, while 82% with $\psi = 1$
312 had reached the same threshold (Supplementary Fig. XX). Similarly, by day 33, 58% of measles epidemics
313 with $\psi = 0$ had reached 5% prevalence *vs.* 85% with $\psi = 1$. The effect was strongly modulated by R_0 : for
314 influenza ($R_0 = 2$), differences across ψ -values were minimal.

315 **Growth rate estimation uncertainty**

316 We simulated 5,000 epidemics in an infinite population for each benchmark pathogen, ran each for 7 days
317 after reaching 100 cumulative cases, binned infections into daily counts, and estimated r via a Poisson GLM
318 with log link. Punctuated individual infectiousness profiles systematically increased variability in $\hat{r}_{MLE}$.

319 [Specific numerical results to be added.]

320 **Generation interval estimation and coverage**

321 We simulated the offspring of 100 index cases, introduced Gamma-distributed observation noise ($a_{obs} = 4$,
322 $b_{obs} = 1$; mean and variance of 4 days), and extracted serial intervals. Posterior distributions for α and β
323 were estimated treating observations as independent. Posterior coverage (95% credible interval containing
324 the true value) decreased dramatically as the individual infectiousness profile narrowed.

325 [Specific coverage numbers to be added.]

326 **Detailed overdispersion results**

327 To capture the relationship between contact variability (c_{amp} or σ), ψ , and OD, we simulated the offspring
328 of 10,000 index cases for each benchmark pathogen and both contact processes across a range of ψ , c_{amp} ,
329 and σ values. We estimated OD by moment-matching the mean and variance of the offspring counts to
330 a Negative Binomial distribution. Overdispersion was highest for narrow infectiousness profiles ($\psi \rightarrow 0$)
331 and high contact variability.

332 Extinction probabilities from 1,000 finite-population epidemics per scenario were systematically higher
333 for more punctuated individual infectiousness profiles.

334 **Detailed D&I intervention results**

335 For symptom-based detection, TE dropped as the typical detection time shifted from before to after peak
336 infectiousness, with the sharpest drops for narrow infectiousness profiles. When the typical detection time
337 was near the time of peak infectiousness, the absolute discrepancy in TE across ψ -values was substantial.

338 [Specific numerical results to be added.]

339 For test-based screening, similar patterns were observed with testing interval replacing symptom offset
340 as the key parameter.

341 **Gathering size restriction results**

342 [Detailed gathering size restriction simulation results to be added, including specific values of σ , c_{max} , and
343 the comparison between gathering-size-restricted and R_{eff} -matched epidemic simulations.]

Supplementary Figures

[Supplementary figures to be inserted here. Expected figures include:

- Supplementary Fig. 1: Overdispersion as a function of ψ under stochastic contacts with individual-level variation.
- Supplementary Fig. 2: Epidemic establishment time distributions across ψ -values for each benchmark pathogen.
- Supplementary Fig. 3: Growth rate estimation variability across ψ -values.
- Supplementary Fig. 4: Generation interval estimation coverage as a function of ψ .
- Supplementary Fig. 5: Effective sample size n_{eff} as a function of ψ and cluster size.
- Supplementary Fig. 6: Testing effectiveness surfaces (TE as a function of detection timing and ψ).
- Supplementary Fig. 7: Truncated generation interval distributions under D&I for different ψ -values.
- Supplementary Fig. 8: Overdispersion introduced by D&I interventions as a function of ψ and detection timing.
- Supplementary Fig. 9: Power analysis heatmaps for ψ inference.
- Supplementary Fig. 10: Gathering size restriction effects on extinction probability.

]

Supplementary Tables

[Supplementary tables to be inserted here. Expected tables include:

- Supplementary Table 1: Fixed generation interval parameters used for empirical ψ estimation, by pathogen.
- Supplementary Table 2: Summary of transmission-tree data: number of clusters, serial intervals, and cluster sizes per pathogen.
- Supplementary Table 3: Epidemic extinction probabilities across ψ -values and contact scenarios for each benchmark pathogen.

]

References

Louis Yat Hin Chan, Sinead E. Morris, Melissa S. Stockwell, Natalie M. Bowman, Edwin Asturias, Suchitra Rao, Karen Lutrick, Katherine D. Ellingson, Huong Q. Nguyen, Yvonne Maldonado, Son H. McLaren, Ellen Sano, Jessica E. Biddle, Sarah E. Smith-Jeffcoat, Matthew Biggerstaff, Melissa A. Rolfes, H. Keipp

373 Talbot, Carlos G. Grijalva, Rebecca K. Borchering, and Alexandra M. Mellis. Estimating the generation
 374 time for influenza transmission using household data in the United States. *Epidemics*, 50:100815, March
 375 2025. ISSN 17554365. doi: 10.1016/j.epidem.2025.100815. URL [https://linkinghub.elsevier.com/
 376 retrieve/pii/S1755436525000039](https://linkinghub.elsevier.com/retrieve/pii/S1755436525000039).

377 Don Klinkenberg and Hiroshi Nishiura. The correlation between infectivity and incubation period of
 378 measles, estimated from households with two cases. *Journal of Theoretical Biology*, 284(1):52–60, Septem-
 379 ber 2011. ISSN 00225193. doi: 10.1016/j.jtbi.2011.06.015. URL [https://linkinghub.elsevier.com/
 380 retrieve/pii/S0022519311003146](https://linkinghub.elsevier.com/retrieve/pii/S0022519311003146).

381 Alun L. Lloyd. Realistic Distributions of Infectious Periods in Epidemic Models: Changing Patterns of
 382 Persistence and Dynamics. *Theoretical Population Biology*, 60(1):59–71, August 2001. ISSN 00405809. doi:
 383 10.1006/tpbi.2001.1525. URL <https://linkinghub.elsevier.com/retrieve/pii/S0040580901915254>.

384 Mattia Manica, Alfredo De Bellis, Giorgio Guzzetta, Pamela Mancuso, Massimo Vicentini, Francesco Ven-
 385 turelli, Alessandro Zerbini, Eufemia Bisaccia, Maria Litvinova, Francesco Menegale, Carla Molina Grané,
 386 Piero Poletti, Valentina Marziano, Agnese Zardini, Valeria d’Andrea, Filippo Trentini, Antonino Bella,
 387 Flavia Riccardo, Patrizio Pezzotti, Marco Ajelli, Paolo Giorgi Rossi, and Stefano Merler. Intrinsic gener-
 388 ation time of the SARS-CoV-2 Omicron variant: An observational study of household transmission. *The
 389 Lancet Regional Health - Europe*, 19:100446, August 2022. ISSN 26667762. doi: 10.1016/j.lanepe.2022.100446.
 390 URL <https://linkinghub.elsevier.com/retrieve/pii/S2666776222001405>.

391 Dylan Morris, John Maclean, and Andrew J. Black. Computation of random time-shift distributions
 392 for stochastic population models. *Journal of Mathematical Biology*, 89(3):33, September 2024. ISSN
 393 0303-6812, 1432-1416. doi: 10.1007/s00285-024-02132-6. URL [https://link.springer.com/10.1007/
 394 s00285-024-02132-6](https://link.springer.com/10.1007/s00285-024-02132-6).

395 Helen J Wearing, Pejman Rohani, and Matt J Keeling. Appropriate Models for the Management of Infectious
 396 Diseases. *PLoS Medicine*, 2(7):e174, July 2005. ISSN 1549-1676. doi: 10.1371/journal.pmed.0020174. URL
 397 <https://dx.plos.org/10.1371/journal.pmed.0020174>.